Assessing the Techno-Economic Feasibility of Solvent-Based, Critical Material Recovery from Uncertain, End-of-Life Battery Feedstock



Chukwunwike O. Iloeje, Yusra Khalid, Joe Cresko and Diane J. Graziano

Abstract As emerging technologies drive up demand for rare earths, value recovery from recycling end-of-life products provides an option for partially closing the material loop, conserving natural capital and enhancing resource security. Yet the techno-economic feasibility of recycling depends on uncertainties associated with the feed input to the recovery process, and the effect of these uncertainties on the viability of the recycling facility. In this study, we couple a first-principle solvent extraction model with an economic model for a separation facility and apply it to assess byproduct recovery and rare earth separation from spent nickel-metal hydride batteries, illustrating the significance of parametric uncertainties. The study shows the importance of risk-informed decision making in the investment, design, and operation of recycling facilities.

Keywords Rare earth · Critical material recovery · NiMH battery recycling · Solvent extraction · Stochastic optimization · Uncertainty analysis · Gibbs energy minimization

The manuscript was written with contribution from all authors.

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X. Chen et al. (Eds.), *Energy Technology 2020: Recycling, Carbon Dioxide Management, and Other Technologies*, The Minerals, Metals & Materials Series,
https://doi.org/10.1007/978-3-030-36830-2_26

Introduction

Rare earth elements (REEs) play a critical role in emerging technologies that span electronics, clean energy, and transportation [1–4]. While recent trends in the evolution of these technologies drive up their demand, current supply chains and end-of-life disposal create non-trivial economic and security concerns. China's dominance in rare earth production increases supply risk [5]. Single-pass use strains primary resources [6] and increases landfill disposal, with accompanying loss of embodied value. Recycling end-of-life products can enhance supply security, conserve natural capital, and minimize landfill disposal [7–9].

Yet the techno-economic viability of recycling depends on compositional and volumetric uncertainties associated with the feed input to the recovery process, their effect on recycling facility economics, and consequent environmental implications. A specific challenge for battery recycling comes from the fact that different manufacturers develop different battery chemistries (material compositions), and these chemistries have evolved over time [10, 11]. Therefore, the pool of batteries available for recycling will have varying compositions, and the decision to build a facility must be made without certainty on the actual feed compositions and with limited knowledge of available battery volumes for recycling.

This paper assesses the techno-economics of recycling—with consideration for resulting environmental impacts—while accounting for uncertain battery feed composition and volume. A few prior studies that looked at the economics of REE recycling from magnets, batteries, or other electronics, addressed scale, and acknowledged the importance of uncertain REE composition without quantifying it [12–14]. This study's key contribution is to quantify the impact of this uncertainty by treating composition as a stochastic parameter in the model and propagating its effect across the recycling facility design and economics. We also analyze the effect of scale for insights on how the available feed volume may affect recycling feasibility assessment. We illustrate this approach for spent nickel-metal hydride (NiMH) battery recycling from hybrid electric vehicles (HEVs) because of their importance to future clean energy security and their current dependence on critical REEs. We limit compositional variability to the REE components of the battery materials, and analyze an integrated battery recycling plant comprising a separation process for byproduct recovery, and solvent extraction for REE separation. We use this analysis to answer the question of how scale and uncertain composition affect recycling.

Method

We set up our study to capture the uncertainty in composition and available quantity of end-of-life NiMH batteries on the techno-economic feasibility of recycling. NiMH batteries contain a number of components, including base metals—nickel (Ni), cobalt (Co), manganese (Mn)—and REEs—neodymium (Nd), praseodymium (Pr), cerium

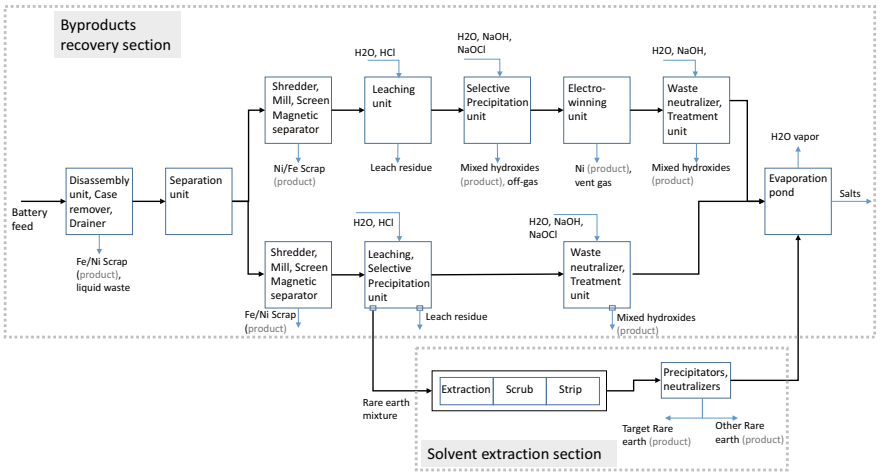


Fig. 1 Flowsheet of overall battery recycling process showing the byproduct recovery and solvent extraction sections

(Ce), and lanthanum (La) [10, 11]. Since this study focuses on the variability in only the REE compositions, we consider all other recovered materials as byproduct. Following this logic, we break up the recycling facility model into the byproduct recovery section and solvent extraction section (for REE separation). Figure 1 shows a high-level representation of the combined recycling process.

Byproduct Recovery Process

The byproduct recovery section is based on the "physical separation/chemical process" configuration described by Sabatini et al. [11] Each box in Fig. 1 represents a conceptual grouping of operations, which may take place in several equipment. The overall process comprises a physical separation part (from disassembly to magnetic separation), a chemical separation part (from leaching to electrowinning, or solvent extraction), and a waste treatment stage (from waste neutralization to evaporation). The physical part primarily recovers Ni and Fe scrap, while the chemical and waste neutralization parts recover pure Ni, other metallic hydroxides and REE chlorides (see Sabatini et al. for a detailed process description. We have also provided an example datasheet with detailed material flows as part of the supplemental material, SI-A1). For the techno-economics of this section, we used a material balance model scaled from their process configuration and material flow data [11]. The material balances of the physical separation processes were determined by applying separation efficiencies derived from their material data table. For the chemical separation and waste neutralization parts, the material flows were determined by balancing the

overall chemical reactions in each equipment, and where applicable, applying conversion efficiencies derived from same reference. The resulting model was set up to scale with the battery feed rate.

Solvent Extraction

The solvent extraction section (Fig. 1) includes the primary separation module comprising the extraction, scrub and strip column stages, as well as associated equipment like precipitators and neutralizers. Here, the REE mixture from the preprocessing section can be separated into REE groups or individual REE streams, depending on the design target and process configuration [15]. The extraction column stages enable the transfer of target REEs into the extractant stream; the scrub column increases the purity of the extracted component(s) by displacing the other components (in this case, Ce and La) from the extractant, and the strip column recovers the high purity product. The design in this study uses a single module to separate the feed into REE mixtures of Nd, Pr and Ce, La (Figure S1-B1). Two main considerations influenced this choice: (1) the prevalence of Nd, Pr mixtures in batteries and in high performance NdFeB magnets [2], and (2) minimal economic incentive for further separating Ce from La, given the associated cost and low product value [15]. The solvent extraction model used in this study was described in our earlier work. [15].

Optimization

Since one of our key objectives is to understand the effect of uncertain REE composition in the feed, the process optimization focuses entirely on the solvent extraction section. The overall stochastic optimization problem minimizes the expected value of a cost function, $\mathrm{E}\left(f\left(x^i\right)\right)$, given a distribution of uncertain input feed compositions, and subject to specified equality $h\left(x^i\right) = 0$ and inequality $g_k\left(x^i\right) = 0$ constraints. Choices for the cost function include the total number of solvent extraction stages, capital costs, and operating costs. The equality constraint is the physics-based constraint representing the set of Gibbs minimization equations for each column stage—embedded in the simulation model—with implicit mass, species, elemental, and charge balances [15]. The inequality constraints target product purity and recovery specs. Table SI-B1 (supplemental material) lists the operating and design optimization variables, x^i, with their respective bounds, while Table SI-B2 shows the recovery and purity specs. Our actual implementation follows the alternating direction method of multipliers algorithm (ADMM) [16] but adapted for optimal consensus stochastic optimization (Algorithm 1).

Algorithm 1: Optimal consensus via ADMM

Step	Action
1	$\rho \in \mathbb{R}^{n_v}; \lambda \in \mathbb{R}^{n_s \times n_v}; \varepsilon \in \mathbb{R}; x \in \mathbb{R}^{n_s \times n_v}; \bar{x} \in \mathbb{R}^{n_v}; i = 1, 2, \ldots, n_v; j = 1, 2, \ldots, n_s; k = 1, 2, \ldots, n_c$
2	$\hat{x}_j^i \leftarrow \underset{x^i}{\operatorname{argmin}}\left(f\left(x_j^i\right) + \varphi\left(h\left(x_j^i\right)\right) + \sum_k p_k\left(\max\left(0, g_k\left(x_j^i\right)\right)\right)\right)$
3	$if \left\|\max(\hat{x}_j^i) - \min(\hat{x}_j^i)\right\| < \varepsilon$, Stop
4	$\bar{x}^i = \frac{1}{n_s}\sum_j^{n_s} \hat{x}_j^i$
5	$\lambda_j^i = \lambda_j^i + \rho_j^i\left(x_j^i - \bar{x}_j\right) \forall i, j$
6	$\hat{x}_j^i \leftarrow \operatorname{argmin}_{x^i}\left(f\left(x_j^i\right) + \varphi\left(h\left(x_j^i\right)\right) + \sum_k p_k\left(\max\left(0, g_k\left(x_j^i\right)\right)\right) + \sum_{i=1}^{n_v} \lambda_j^i\left(x_j^i - \bar{x}^i\right) + \sum_i^{n_v} \frac{\rho_j^i}{2}\left(x_j^i - \bar{x}^i\right)^2\right)$
7	Go back to step 3

Here, ρ, λ are the linear and quadratic stochastic penalty parameters, ε is the convergence threshold, $x, \hat{x}, \bar{x}$ represent the process variables (Table SI-B1), their optimal values, and the mean of the optimal values across all sampled compositions, and n_v, n_s, and n_c represent the number of optimization variables, number of samples, and number of inequality constraints, respectively. Step 2 reformulates the standard optimization problem by moving the constraints into the objective. The first term is the original objective function; the second term represents the requirement that the physics-based Gibbs model be satisfied for the evaluation to be feasible. The third term represents an exact penalty function method that penalizes any inequality constraint violation [17, 18]—in this case, deviations from purity and recovery specs. Step 3 tests for convergence and stops the program if it meets the convergence criteria. Steps 4 and 5 update $\bar{x}$ and λ at each iteration, and Step 6 represents the augmented objective function. Algorithm 1 iteratively drives the solution towards a consensus optimum over the space of compositions.

Economic Model

To determine the battery recycling process economics, we adopt a bottom-up approach that takes all material flows as input, then proceeds to evaluate the corresponding capital costs, operating costs, and revenue.

Capital equipment cost: For the capital cost, we size the individual equipment, evaluate corresponding equipment cost, and then apply capital project econometric factors to determine the required total capital investment (C_{TCI}) [19]. Equation 1 represents the general formulation for the capital equipment cost (C_{capex}), where C_{ref_i} is the reference cost of equipment i [11]; $\left(\frac{m}{m_{\text{ref}}}\right)^n$ is the scaling parameter [19, 20] based on the ratio of actual to reference mass or volumetric flow rates; and the PPI() term uses the producer price index to convert prices to the project's current year equivalent. The total capital investment (TCI) is determined using Eq. 2, where f_{directs} account for costs like installation and instruments and controls, and $f_{\text{indirects}}$ include supervision, legal, and contingency. [19] See tables SI-C1 and SI-C2 for parameter values used.

$$C_{\text{capex}} = \sum_i \left(C_{\text{ref}_i} \left(\frac{m_i}{m_{\text{ref}_i}} \right)^n * N_{\text{units}} * f_{\text{mat}} * \text{PPI}\left(\frac{2018}{\text{Year}_{\text{ref}}} \right) \right) \tag{1}$$

$$C_{\text{TCI}} = C_{\text{capex}} \left(1 + f_{\text{delivery}}\right) \left(1 + \sum f_{\text{directs}} + \sum f_{\text{indirects}}\right) \tag{2}$$

Operating equipment cost: Operating costs include the cost of utilities, raw materials, labor, and other miscellaneous costs (Tables SI-C3 to SI-C5) determined by summing up all these contributions (Eq. 3). Here, m_j represents the relevant quantity (e.g., solvents, process water), C_j represents the cost per unit quantity, f_k the multiplier factor for each miscellaneous cost item (e.g., supervision, taxes), and

C_k is their associated cost basis (e.g. fixed capital, annual labor costs).

$$C_{\text{opex}} = \sum_k f_k C_k + \sum_j m_j C_j \tag{3}$$

Revenue and Net Present Value (NPV): The recycled product groups considered in this study include the byproduct—Ni/Fe scrap, Fe/Ni scrap, nickel metal, hydride alloy scrap—and REE products. The revenue equation is shown in Eq. 4, where m_i represents the quantity of recovered product (or byproduct) i, and p_i represents the price per unit of product (or byproduct) i, as detailed in Tables SI-C6 and SI-C7. NPV is evaluated using Eq. 5, where C_t is the net cash flow $\left(C_{\text{rev}} - C_{\text{opex}}\right)$ for a given period ($t = 1$ year), T is the financial life of the recycle facility, and r is the expected rate of return.

$$C_{\text{rev}} = \sum_i m_i p_i \tag{4}$$

$$\text{NPV} = \sum_{t-1}^{T} \frac{C_t}{(1+r)^t} - C_{\text{TCI}} \tag{5}$$

Model Integration

Figure 2 illustrates the model integration logic for the battery recycling facility, highlighting how the different models interact and exchange information. We start with a default composition of battery feed shown in Table 1.

For the purposes of this study—which focuses on the variation in REE compositions—we assume that the overall battery composition in Table 1 remains fixed,

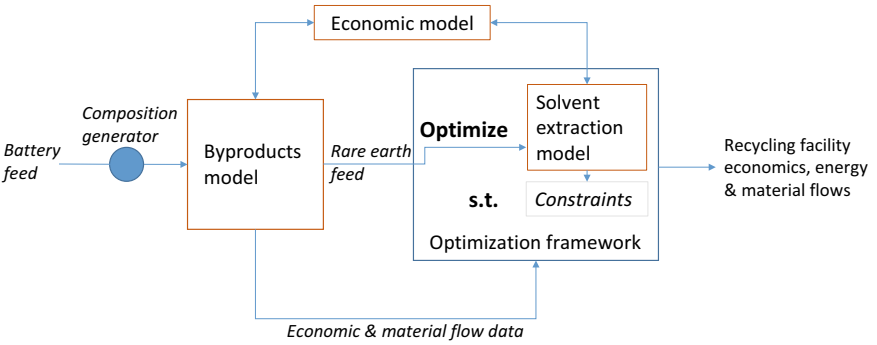


Fig. 2 Model integration logic

Table 1 NiMH battery composition

Ni	Fe	Rare earths	Co	Mn	Al	KOH	H_2O	Leveling agents	Plastic	$Ni(OH)_2$
0.213	0.435	0.050	0.017	0.008	0.003	0.030	0.060	0.010	0.050	0.125

Derived from data in [11]

Table 2 Relative REE compositions for generating feed distribution

	$LaCl_3$ (%)	$CeCl_3$ (%)	$PrCl_3$ (%)	$NdCl_3$ (%)
Composition	35–40	12–40	5–11	20–38

Derived from data in [21, 22]

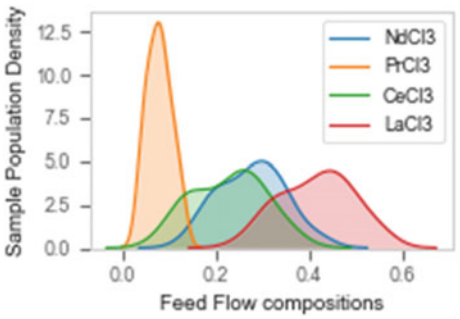


Fig. 3 Kernel density distribution of feed concentration samples for Nd, Pr, Ce, and La in battery feed. Kernel density estimate represents the probability density function of a random variable, in this case, feed composition

while the relative composition of the individual REEs varies. To capture this variation, we select a reference composition range from the literature (Table 2) [21, 22]. We pass the information in Table 2 to the composition generator that uses the Latin hypercube approach [23] to sample the REE composition space (Fig. 3). Each sample composition is combined with the data in Table 1 and fed into the byproduct recovery model. The REE mixture recovered from the byproduct section proceeds to the solvent extraction model embedded in the optimization framework. Both the solvent extraction and byproduct recovery models interact with the economic model to predict capital costs, operating costs, and other relevant economic indicators. See supplemental material (SI-B) for implementation details.

Results

Effect of Compositional Variation

The spread of the distribution in Fig. 4 highlights the strong dependence of the solvent extraction system size on variability in feed composition. It shows the distribution of predicted minima for total number of the solvent extraction column stages required to separate the REE groups given the sampled feed compositions. The dotted line marks the result of our phase-1 stochastic optimization (using Algorithm 1) and corresponds to a consensus configuration (nine equilibrium extraction stages, seven scrub stages and two strip stages) over the space of sampled compositions that meet the purity (99%) and recovery (as high as possible) specifications.

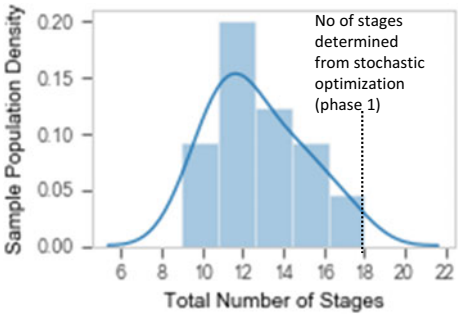


Fig. 4 Distribution of predicted number of solvent extraction stages given the distribution of feed compositions

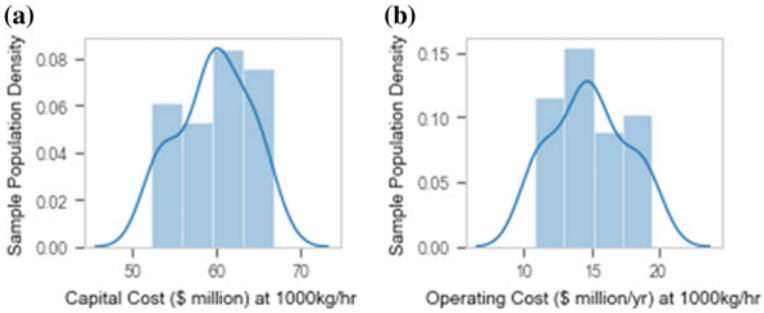


Fig. 5 Distribution of **a** capital cost and **b** operating cost for reference case with 1000 kg/h battery feed rate

Using this 18-stage configuration, we proceed to phase-2, where we minimize the sum of the capital and operating costs of the recycling facility for 36 different feed compositions, with the fractions sampled from the distribution in Fig. 3. Figure 5 summarizes the result of this optimization. The distributions show standard deviations of about 7% for the capital cost and nearly 20% for the annual operating cost of the reference plant (sized for 1000 kg/h or 5 batteries/h). This spread is significant, especially since recycling facilities tend to have low-to-moderate rates of return.

Figure 6 provides some insight into the relationship between the cost estimates and feed compositions. In general, there appears to be a positive, linear correlation between the fraction of the target REEs (in this case, Nd and Pr) and both the capital and operating costs. This is probably due to the larger extractant flow rates required to support higher component loading and the subsequently larger flowrates of scrubbing and stripping streams, all of which require larger equipment sizes and increased startup cost. Note that these are aggregate behaviors, as the data points in Fig. 6 show that individual cases can trend differently.

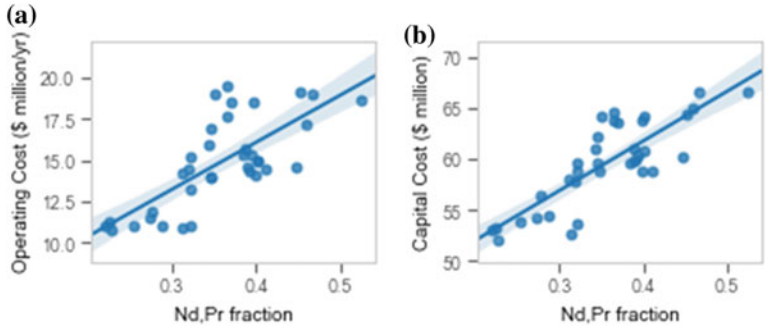


Fig. 6 Variation of **a** total facility capital costs, and **b** total facility operating costs with target (Nd, Pr) fraction in the REE mixture at reference scale (1000 kg/h battery feed). The dots represent individual data points for each sample; the line is a regression fit, and the shaded area visualizes the confidence interval around the regression line

Effect of Scale (Battery Feed Volume)

The other factor we examined was the effect of scale—specifically, the quantity of batteries available to the facility for recycle. We selected ten recycling facility sizes, ranging from feed rates of one battery per hour (200 kg/h) to nineteen batteries per hour (3800 kg/h), and we evaluate these scales for each of the sampled compositions. Figure 7 shows the regression plots for the variation of capital and operating costs with feed amount. Each point on the graph represents the mean of a spread of values at each scale. Our choice of a linear estimator to represent the trend lines is solely for visualization purposes, as it only approximates the trend in the underlying data.

To complete the analysis of scale, we consider how our results relate to the minimum sustainable price (MSP) of a target product from the recycling plant. Since this study is interested in understanding the economics of recovering critical REE materials as ready-to-market products from the recycling facility, we rephrase the question of scale as a hypothesis about what target REE price point will make such a facility economically viable, and how this price evolves with scale. We select Nd, Pr as the target REEs for which to determine the MSP, and combine Eqs. 4 and 5 to evaluate the corresponding MSP at for each sample. Figure 8 summarizes the answer to the MSP question as posed in this study. Besides showing the strong dependence of MSP on scale, it also highlights the minimum recycling facility size required to entertain hopes of economic viability.

Discussion

Our analysis shows that uncertain REE composition in battery feed significantly affects the capital and operating economics of recycling, and as such should be a key

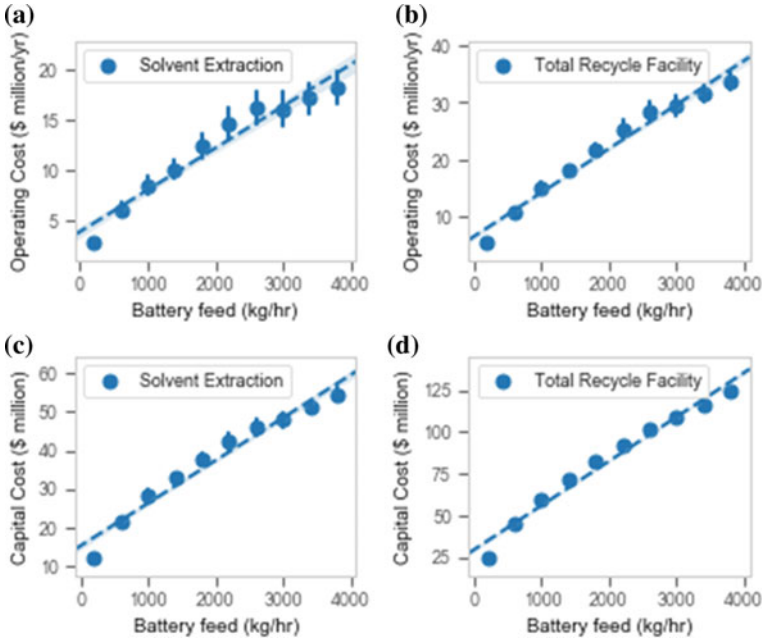


Fig. 7 Capital and operating costs as a function of battery feed rate (scale). **a** and **c** show the relevant costs for the solvent extraction section (for REE separation), while **b** and **d** show those for the entire facility. The dots represent sample means at each scale, and the shaded region visualizes the associated confidence interval

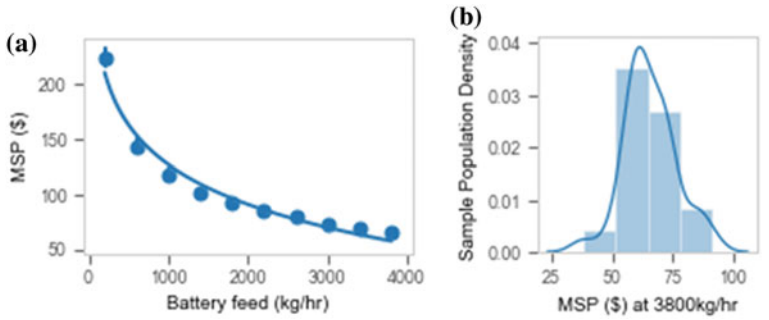


Fig. 8 Minimum sustainable price (MSP) of Nd, Pr chloride **a** as a function of scale—each circle shows the mean MSP at the corresponding battery feed rate, and **b** MSP distribution at 3800 kg/h feed

component of risk analysis prior to a decision to build and operate the facility. Factoring this uncertainty into the design of the facility will reduce the associated capital investment risk. Our results also show that scale matters (Fig. 8), and the prospect of economic viability of battery recycling increases proportionally to the volume of

batteries available for recycling. Based on our estimates, processing 27,000 t per year (136,000 batteries) puts the mean MSP at about $65/kg for the Nd, Pr chloride product, 40% above prevailing market price for didymium oxide and 10% above that for didymium. This makes a strong case for upgrading the product to the alloy form, assuming that the higher revenue from Ce, La misch sales (from $1.715/kg to $4.9/kg) will offset the increase in capital cost and lower the resulting mean MSP. For battery chemistries with high La, Pr content [11], one may consider further separating Nd from Pr, to benefit from the higher Pr market price ($97/kg). The flip side of the volume problem relates to the actual amount of HEV NiMH batteries available for recycling. Assuming an average use-phase of 10 years, and that all batteries are recycled at end-of-life, Fig. 9 shows that a single facility (starting operation in 2020) requires about 25–45% of the projected battery volume available by 2020, 27–52% by 2030, and 36–70% by 2040. One can already see a bias towards a single large plant, a conclusion consistent with other studies that have assessed value recovery from recycling at scale in the USA [14]. In reality, only a fraction of those batteries will be available for recycling—recycling faces stiff competition from alternative end-of-life pathways, including user retention, dump stockpiling, and refurbishing for reuse. Addressing this challenge of uncertain volumes requires targeted measures to address supply logistics upstream of the recycling plant. Another option is to explore source-agnostic flexible recycling plants that simultaneously handle other battery types (e.g. consumer batteries).

Scaling up the recycling facility also brings with it other less desirable side effects, as the increased recycling facility size also increases its environmental signature. With ever-larger feed volumes, it must deal with larger amounts of toxic effluents like chlorine gas (Fig. 10a), and significant amounts of hydrogen flare (Fig. 10b), the latter of which represents a lost opportunity for energy integration.

Limitations in this analysis include: overlooked economic factors (we did not include the cost of battery collection; on the other hand, we assumed once through

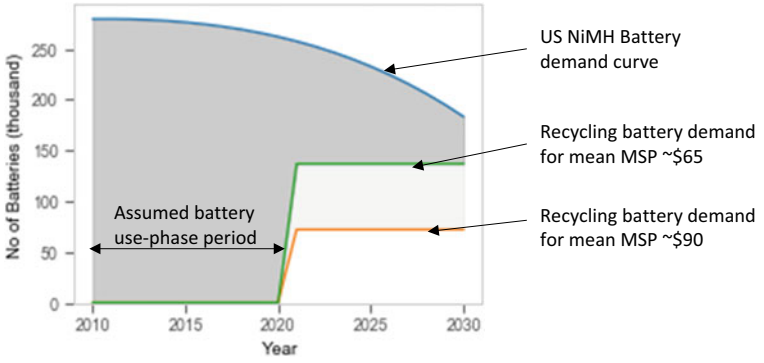


Fig. 9 Projection of NiMH battery demand in the USA, compared to recycling scales needed for mean MSP estimate of $90 (72,000 batteries per year) and $65 (136,800 batteries per year). The demand curve was estimated from data in [24–27]

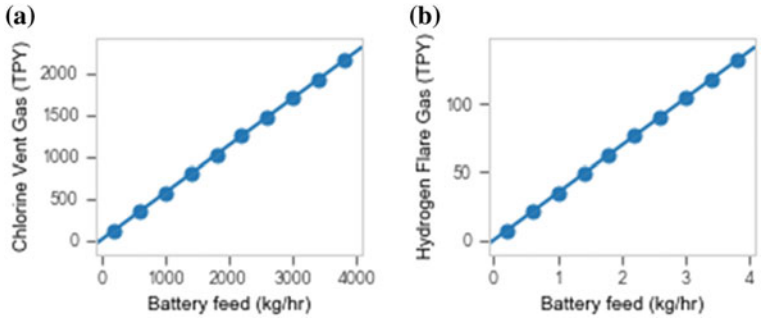


Fig. 10 Estimates of toxic gas emissions from the battery recycling facility. **a** Chlorine vent gas from electrowinning process, and **b** H_2 flare gas, primarily from leaching

acid consumption—no reuse—which over-predicts acid and neutralizer consumption); sub-optimal solutions (we used a non-adaptive scheme to penalize constraint violations, running the risk of finding local optima); and other simplifying assumptions (we neglected the impact of composition variation for non-REE battery materials; we also overlooked product market uncertainty). While these assumptions may affect the absolute values of the reported results, we expect the change to be across board, so the trends revealed in the foregoing analysis remain applicable. Some of the findings are specific to the hydrometallurgical process considered in this study, as well as equipment choices made. Emerging technologies might offer more environmentally sustainable paths to recycling, and possibly support economic, small-scale distributed recycling[28, 29].

Conclusion

In this study, we analyzed the effect of uncertainties in material composition and available quantities for recycling on the techno-economics of recycling HEV batteries—with consideration for the consequent environmental implications. Given that uncertainty about the volume and material composition of batteries arriving at any proposed recycling facility is a core uncertainty associated with recycling, these results have direct implications on the potential exposure to financial risk, as well as the environmental impact of constructing and operating such a facility. While we focused on batteries in this analysis, the insights we derived also apply to other recycling, and the foregoing analysis can inform a robust assessment of their techno-economic feasibility given these uncertainties.

Acknowledgements We wish to acknowledge Allinson Santos-Xavier (Argonne National Lab) for providing valuable discussions on optimization approaches. The submitted manuscript has been created by UChicago Argonne LLC, Operator of Argonne National Laboratory (“Argonne”). Argonne

National Laboratory's work was **supported by** the US Department of Energy, Office of Energy Efficiency and Renewable Energy (EERE), under contract DE-AC02-06CH11357. The US government retains for itself, and others acting on its behalf, a paid-up non-exclusive, irrevocable worldwide license in said article to reproduce, prepare derivative works, distribute copies to the public, and perform publicly and display publicly, by or on behalf of the government. The Department of Energy will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan. This research used resources of the Argonne Leadership Computing Facility, which is a DOE Office of Science User Facility.

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